IPOD Unification Extended: Interacting IPODs and Structural Routes to Field-Like Laws

(Draft theory paper; conceptual and mathematical positioning, no experiments)

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Abstract

We extend the IPOD (Input–Process–Output–Data) framework from a single selection–update machine to a minimal *interacting* architecture intended to better reflect real entities that must both preserve identity and adapt. Our central claim is structural, not ontological: many "selection and update" phenomena admit a shared representation as coupled stochastic kernels with state updates. We formalize (i) IPOD machines as kernel–update systems, (ii) a notion of structural equivalence via commuting representation maps, and (iii) a two-engine coupling pattern that we interpret as *stabilization* and *growth*. Under explicit modeling assumptions—in particular, an isotropic Green's-function ansatz for an additive score field and a local-move output space—we show how inverse-square drift fields and Poisson-type continuum structure arise as mean-field limits. We then outline (without claiming full derivations) how additional axioms such as locality, conservation laws and gauge symmetry would be required to connect the framework to Maxwell- and Einstein-like field equations. Quantum measurement is treated operationally via quantum instruments (Born-rule kernel plus post-measurement update), and "softmax" appears as a coordinate representation of discrete kernels rather than a universal physical law. This manuscript is a theoretical positioning draft; it does not present empirical validation.

Keywords. IPOD; structural equivalence; interacting kernels; mean-field limits; inverse-square law; Poisson equation; quantum instruments; attention; framing; anchoring.

Status and Scope (Reader Guidance)

This is a theory/structure draft.

- **What is proved:** structural representation results ("can be written as IPOD"), a softmax coordinate lemma for discrete full-support kernels, and a conditional recovery of inverse-square / Poisson structure under explicit assumptions.
- **What is not proved:** that the universe is "made of softmax," nor full derivations of Maxwell or Einstein equations from IPOD alone, nor any empirical claims.
- **Why it matters:** it provides a disciplined way to connect domains without frame drift: the shared invariant is *selection kernel + state update*, not domain ontology.

1 Introduction

Many domains feature systems that begin with a space of possibilities and end with one expressed outcome: a measurement result, a chosen action, a conscious percept, an adopted frame, a selected token. Across domains, it is tempting to declare these "the same mechanism." That move

often fails under audit because meanings, layers, and frames differ (e.g. quantum foundations vs. machine learning vs. cognitive bias vs. social discourse).

This work adopts a narrower objective: *structural unification*. We ask whether apparently disparate phenomena can be represented by the same abstract operational schema: (i) a probabilistic selection rule over outputs, conditioned on inputs and internal state, and (ii) a state update rule that feeds forward into subsequent selections. This leads to the IPOD machine.

A single IPOD is a minimal unit. Real entities, however, typically exhibit a duality: they preserve identity (stability) while adapting (growth). We therefore extend the framework to interacting IPODs, and study minimal coupling patterns. We then explore a *conditional* route from coupled IPOD networks to field-like structures, clarifying which parts follow from which assumptions.

Contributions.

1. A formal definition of IPOD machines and structural equivalence (representation morphisms).
2. A minimal model of interacting IPODs (two-engine coupling).
3. A softmax coordinate lemma (softmax as representation, not ontology).
4. A conditional derivation: inverse-square drift and Poisson structure arise under an isotropic Green's-function ansatz plus local-move selection.
5. A disciplined "program" section explaining what extra axioms are required to reach Maxwell/Einstein-like equations, without overclaiming derivation.

2 IPOD Machines and Structural Equivalence

2.1 IPOD machine

Definition 2.1 (IPOD machine)**.** An *IPOD machine* is a tuple

$$\mathcal{M} = (\mathcal{I}, \mathcal{O}, \mathcal{D}, K, U)$$

where $\mathcal{I}$ is an input space, $\mathcal{O}$ an output space, $\mathcal{D}$ a data/state space, K a selection kernel

$$K : \mathcal{I} \times \mathcal{D} \to \Delta(\mathcal{O}), \quad (o \mapsto K(o \mid i, d)),$$

and U is a state update map

$$U : \mathcal{D} \times \mathcal{I} \times \mathcal{O} \to \mathcal{D}.$$

The dynamics are

$$O_t \sim K(\cdot \mid I_t, D_t), \qquad D_{t+1} = U(D_t, I_t, O_t). \tag{1}$$

Remark 2.2 (Structural slots vs. domain meaning)**.** The symbols $(\mathcal{I}, \mathcal{O}, \mathcal{D}, K, U)$ are abstract slots. Domain meanings are assigned by interpretation; structural unification concerns the existence of such an assignment, not identity of ontology across domains.

2.2 Structural equivalence

Definition 2.3 (Structure-preserving morphism)**.** Let $\mathcal{M} = (\mathcal{I}, \mathcal{O}, \mathcal{D}, K, U)$ and $\mathcal{M}' = (\mathcal{I}', \mathcal{O}', \mathcal{D}', K', U')$ be IPOD machines. A morphism $\Phi : \mathcal{M} \to \mathcal{M}'$ is a triple of maps

$$\phi_I : \mathcal{I} \to \mathcal{I}', \quad \phi_O : \mathcal{O} \to \mathcal{O}', \quad \phi_D : \mathcal{D} \to \mathcal{D}'$$

such that for all (i, d):

(i) (*Kernel commutation*) the pushforward of $K(\cdot \mid i, d)$ by ϕ_O equals $K'(\cdot \mid \phi_I(i), \phi_D(d))$.

(ii) (*Update commutation*) $\phi_D(U(d,i,o)) = U'(\phi_D(d), \phi_I(i), \phi_O(o))$ for all o.

Remark 2.4. This captures "same structure" operationally: selection distributions and state updates commute under representation maps. It is the formal content of "equivalent IPOD structure."

3 Softmax as a Coordinate Representation

Lemma 3.1 (Softmax representation of full-support discrete kernels)**.** *Assume $\mathcal{O}$ is finite and $K(o \mid i,d) > 0$ for all $o \in \mathcal{O}$. Define $A(o \mid i,d) = \log K(o \mid i,d)$. Then*

$$K(o \mid i,d) = \frac{e^{A(o|i,d)}}{\sum_{o'} e^{A(o'|i,d)}} = \mathrm{Softmax}(A(\cdot \mid i,d))(o).$$

Proof. Substitute $A = \log K$. The denominator is $\sum_{o'} K(o' \mid i,d) = 1$. □

Remark 3.2 (Interpretation)**.** Lemma 3.1 means: "softmax" can be a mere coordinate choice (log-probabilities) for discrete kernels. Therefore, the invariant object is the kernel K, not the presence of softmax in notation.

4 Interacting IPODs: Minimal Two-Engine Coupling

4.1 Coupled machines

Definition 4.1 (Two-IPOD system)**.** A *two-IPOD system* consists of two IPOD machines $\mathcal{M}^{(1)}$ and $\mathcal{M}^{(2)}$ whose kernels and updates may depend on both states:

$$O_t^{(1)} \sim K^{(1)}(\cdot \mid I_t^{(1)}, D_t^{(1)}, D_t^{(2)}), \qquad D_{t+1}^{(1)} = U^{(1)}(D_t^{(1)}, D_t^{(2)}, I_t^{(1)}, O_t^{(1)}), \tag{2}$$

$$O_t^{(2)} \sim K^{(2)}(\cdot \mid I_t^{(2)}, D_t^{(2)}, D_t^{(1)}), \qquad D_{t+1}^{(2)} = U^{(2)}(D_t^{(2)}, D_t^{(1)}, I_t^{(2)}, O_t^{(2)}). \tag{3}$$

4.2 Interpretation: stabilization and growth (as constraints)

Remark 4.2 (A minimal formal reading)**.** The terms *stabilization* and *growth* are interpretive labels, but can be pinned to constraints:

- A stabilization engine tends to preserve certain invariants or reduce deviation from a reference manifold (e.g. boundaries, conservation-like constraints).
- A growth engine tends to explore, adapt, or reduce predictive error under changing inputs, often tolerating higher uncertainty.

This manuscript uses these labels only to motivate coupling, not as a proved physical principle.

5 A Conditional Route to Field-Like Laws

This section is *conditional*. We state explicit assumptions; the results that follow are consequences of those assumptions, not of IPOD alone.

5.1 Assumption set for spatial interaction

Assumption A1 (spatial embedding). Each entity i has a position $\mathbf{r}_i \in \mathbb{R}^3$.

Assumption A2 (local-move output space). The output of a "motion" IPOD is a small move option $o \in \mathcal{O}_i = \{\varepsilon \mathbf{u}_k\}_{k=1}^m$ for unit directions $\mathbf{u}_k$.

Assumption A3 (additive score field). There exists a scalar score field $A_{\text{tot}} : \mathbb{R}^3 \to \mathbb{R}$ produced additively by sources: $A_{\text{tot}} = \sum_j A_j$.

Assumption A4 (isotropic Green's-function ansatz). Each source j contributes

$$A_j(\mathbf{r}) = \frac{\alpha_j}{\|\mathbf{r} - \mathbf{r}_j\|}. \tag{4}$$

This is chosen because $\|\mathbf{r}\|^{-1}$ is the Green's function of the 3D Laplacian; it is a modeling ansatz.

Assumption A5 (softmax selection over moves). The kernel over move options is

$$K(o \mid \mathbf{r}_i, D_i) = \text{Softmax}(S(\cdot \mid \mathbf{r}_i, D_i))(o), \tag{5}$$

with score

$$S(o \mid \mathbf{r}_i, D_i) = \beta_i A_{\text{tot}}(\mathbf{r}_i + o) + S_0(o \mid D_i), \tag{6}$$

where β_i is a sensitivity and S_0 is a baseline.

5.2 Emergent drift aligns with the score gradient

Proposition 5.1 (Small-step drift follows the score gradient)**.** *Assume symmetric move sets $\mathcal{O}_i$ and sufficiently small ε. Then the expected displacement*

$$\Delta\mathbf{r}_i := \mathbb{E}[o \mid \mathbf{r}_i, D_i] = \sum_{o \in \mathcal{O}_i} o\, K(o \mid \mathbf{r}_i, D_i)$$

is aligned with $\nabla A_{tot}(\mathbf{r}_i)$ up to scaling and baseline terms:

$$\Delta\mathbf{r}_i \propto \nabla A_{tot}(\mathbf{r}_i).$$

Sketch. Expand $A_{\text{tot}}(\mathbf{r}_i + o)$ to first order in o and use symmetry of $\mathcal{O}_i$ to cancel odd terms in the partition normalization; the first non-cancelling contribution to $\mathbb{E}[o]$ is proportional to the gradient. A full derivation parallels standard softmax-policy gradient expansions in small-step settings. □

5.3 Inverse-square structure from the $1/r$ ansatz

Corollary 5.2 (Inverse-square drift from a single source)**.** *Under A1–A5 and for a single source j,*

$$\nabla_{\mathbf{r}_i} A_j(\mathbf{r}_i) = -\alpha_j \frac{\mathbf{r}_i - \mathbf{r}_j}{\|\mathbf{r}_i - \mathbf{r}_j\|^3}.$$

Thus the drift direction has the same spatial scaling as Coulomb/Newton forces, conditionally on the Green's-function ansatz.

5.4 Continuum / mean-field limit and Poisson structure

Proposition 5.3 (Poisson structure as continuum limit)**.** *Let sources form a density $\rho(\mathbf{r})$ of strengths α. Define the continuum field*

$$\Phi(\mathbf{r}) = \int_{\mathbb{R}^3} \frac{\rho(\mathbf{r}')}{\|\mathbf{r} - \mathbf{r}'\|}\, d^3\mathbf{r}'.$$

Then (distributionally) $\nabla^2\Phi(\mathbf{r}) = -4\pi\rho(\mathbf{r})$.

Sketch. $\|\mathbf{r}\|^{-1}$ is the Green's function of $-\nabla^2$ in $\mathbb{R}^3$. □

Remark 5.4 (What is and is not obtained)**.** We obtain a Poisson-type structure because we chose the $1/r$ Green's kernel. This is a structural bridge to classical potential theory, not a derivation of electromagnetism or gravity from IPOD alone.

6 Program Toward Maxwell/Einstein (Outline, Not Derivation)

This section is a *programmatic note.* To connect to Maxwell- or Einstein-like equations, one must introduce additional structure beyond the scalar field Φ.

6.1 What extra axioms are needed

(P1) Gauge structure. To reach Maxwell equations, introduce vector potentials and gauge symmetry (invariance under $A_\mu \mapsto A_\mu + \partial_\mu \chi$). This requires outputs and states that transform covariantly, not just a scalar score.

(P2) Conservation laws. Charge/current conservation (continuity equation) must be enforced as a constraint on updates U across interacting IPODs.

(P3) Relativistic locality. To reach field propagation with finite speed, incorporate spacetime causal structure, which constrains the permissible kernel dependencies.

(P4) Tensor geometry. To approach Einstein equations, the "field" must be metric-like (a symmetric tensor), and updates must respect diffeomorphism invariance.

Remark 6.1. The present paper does not claim these emerge automatically. It clarifies where they would enter in an IPOD-based construction.

7 Quantum Mechanics in the IPOD View (Operational)

7.1 Quantum measurement as IPOD

Quantum measurement admits a rigorous operational representation via quantum instruments.

Proposition 7.1 (Quantum instrument as IPOD)**.** *Let ρ be a density operator (data/state), let M be a measurement setup specified by a POVM $\{E_k\}$ (input), and let outcomes be $k \in \mathcal{O}$ (output). Define*

$$K(k \mid M, \rho) = \mathrm{Tr}(E_k \rho), \qquad U(\rho, M, k) = \frac{\sqrt{E_k}\rho\sqrt{E_k}}{\mathrm{Tr}(E_k \rho)}.$$

Then $(\mathcal{I}, \mathcal{O}, \mathcal{D}, K, U)$ is an IPOD machine.

Remark 7.2 (Softmax and Born rule)**.** If outcomes are discretized and full-support holds, Lemma 3.1 implies one may write K in softmax coordinates with score $A = \log K$. This is a coordinate representation of the kernel, not an algorithmic identity between quantum measurement and transformer attention.

8 Domain Instantiations (Structural, Audit-Safe)

We summarize how different domains fit the IPOD slots (not as identity claims):

- **Transformer generation:** input = token prefix; data = parameters/cache; kernel = softmax over logits; update = append/update cache.
- **Anchoring:** input includes anchor; data = beliefs/memory; kernel = choice/estimate distribution with anchor-distorted score; update = priming/consolidation.
- **Framing:** input = event/discourse; data = cultural/institutional state; kernel = distribution over frames; update = institutionalization and reinforcement.
- **Quantum measurement:** input = instrument; data = density operator; kernel = Born rule; update = instrument map.

9 Limitations and Empirical Agenda

9.1 Limitations

- Structural representability does not imply mechanistic identity.
- The inverse-square/Poisson results depend on explicit modeling assumptions (A1–A5).
- Maxwell/Einstein connections are outlined as a program requiring extra axioms.

9.2 Empirical agenda (future work)

This framework becomes testable when one specifies measurable proxies for:

1. the kernel $K(\cdot \mid I, D)$ (choice probabilities, outcome frequencies, token probabilities),
2. the update U (state transitions, memory effects, institutional dynamics),
3. and the coupling structure among interacting IPODs.

Empirical validation would require domain-specific experimental or observational designs. This manuscript does not present such evidence.

10 Conclusion

We extended the IPOD framework from a single machine to interacting machines, motivated by the stabilization–growth duality observed in real entities. We formalized IPOD dynamics, structural equivalence, and clarified softmax as a coordinate representation of discrete kernels. Under an explicit isotropic Green's-function ansatz and local-move selection, we showed how inverse-square drift fields and Poisson-type structure arise as mean-field limits. We then separated what is structurally recovered from what would require additional axioms to reach full Maxwell/Einstein-type theories. The result is a disciplined unification claim: many domains share a selection–update structure, and certain classical field-like patterns appear under clear assumptions.

Acknowledgments

This manuscript is a draft intended to fix definitions, avoid frame drift, and position a structural program. No empirical claims are made.

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